

UAV Trajectory Optimization for Time-limited Data Collection in Dynamic Wireless Environments

Farzad Mehri, Reza Jahani

Department of Electrical and Computer Engineering, North Carolina State University

{fmehri, rjahani}@ncsu.edu

Abstract—This project investigates UAV trajectory optimization for time-limited multi-base-station (BS) data collection in LTE/5G environments using the AERPAW Digital Twin. Building upon our earlier baseline—where a fixed trajectory was paired with either Grid Search based or MILP-based optimal BS selection—we develop an *adaptive trajectory planning* method that jointly optimizes the visiting order of BS locations and the corresponding BS selection decisions. Starting near LW1, the UAV evaluates all permutations of visiting LW2–LW4, and for each candidate trajectory, computes the optimal BS selection along the path by solving the previously formulated Mixed Integer Linear Program (MILP). A grid search is then performed across all candidate waypoint sequences, and the trajectory with the maximum AADM score is selected. We further evaluate our approach under multiple data demand distributions (uniform, exponential, and random) and visualize the derived trajectories and their associated link-quality characteristics. Simulation results confirm that adaptive trajectory selection, combined with optimal BS assignment, significantly improves achievable scores and demonstrates consistent robustness across varying demand scenarios, outperforming fixed-trajectory baselines.

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs) are increasingly incorporated in wireless networks as airborne vehicles to provide resilient and adaptive wireless connectivity in scenarios such as emergency response, rural communications, and infrastructure inspection. *AERPAW Digital Twin* platform has been recently used by many researchers in which UAVs can interact with other fixed or mobile RF transceivers such as Unmanned Ground Vehicles (UGV) and Base Stations (BS) in a simulated wireless communication system, capturing realistic channel variations and practical scenarios.

This project investigates the problem of **trajectory optimization** for a UAV tasked with downloading data from multiple BSs within a predefined time limit. The objective is to maximize the total downloaded data while maintaining geofence boundaries and landing within the deadline [1].

In this final stage of the project, we implemented and validated the **baseline methods** and the new **Adaptive Trajectory and BS selection**. Specifically, we:

- Developed an **adaptive trajectory planning** framework in which the UAV, starting near LW1, evaluates all possible permutations of visiting the remaining BS locations (LW2, LW3, LW4) along with establishing a baseline method which involves a fixed trajectory.
- For each candidate visiting order, we executed the previously developed optimization algorithm by solving the

Mixed Integer Linear Program (MILP) to determine the *optimal BS selection* at every time step along the trajectory.

- Applied a **grid search** across all candidate waypoint sequences, selecting the trajectory that achieves the maximum AADM score based on joint trajectory and BS-selection performance.
- Performed a binary search on duration of the UAV flight time in order to find the minimum time resulting in the maximum score.
- Conducted extensive simulations under multiple data-demand scenarios (uniform, exponential, and random) to evaluate the effectiveness of the proposed adaptive method.
- Visualized the results through (i) UAV trajectory plots for each candidate and selected path, (ii) total score per data-demand scenario, and (iii) instantaneous SNR profiles observed from each BS.

These extended experiments and adaptive trajectory design results are built upon our baseline evaluation and provide a more comprehensive performance benchmark for optimization aided UAV trajectory design.

II. SYSTEM MODEL

The UAV is modeled as a point-mass system in 3D space, represented by its state:

$$p(t) = [x(t), y(t), z(t), \phi(t), \theta(t), \psi(t)], \quad (1)$$

where (x, y, z) denote the UAV coordinates, and (ϕ, θ, ψ) represent roll, pitch, and yaw, respectively. The UAV's motion is constrained by a maximum velocity:

$$\|r(t+1) - r(t)\| \leq V_{\max}, \quad (2)$$

and altitude/geofence boundaries:

$$z_{\min} \leq z(t) \leq z_{\max}, \quad r(t) \in \mathcal{G}. \quad (3)$$

The downlink channel between the UAV and BS i provides an instantaneous signal-to-noise ratio (SNR) $\gamma_i(t)$, determined by the UAV's position and a deterministic two-ray ground-reflection model:

$$\gamma_i(t) = \frac{P_i G_i}{N_0 L_i(r(t))}, \quad (4)$$

where P_i is the transmit power, G_i is the antenna gain, N_0 is the noise power, and $L_i(\cdot)$ represents path loss. The achievable

data rate is retrieved from a lookup table as $R_i(t) = f(\gamma_i(t))$ (Mbit/s).

Each BS BS_i announces the amount of data it has, C_i , at the beginning of the experiment. Total demand of the BSs can be shown as $C_T = \sum_i(C_i)$

The control variables at each step t are:

$$a(t) = \{p_{t+1}, i^*(t)\},$$

where p_{t+1} is the next waypoint and $i^*(t)$ is the selected BS. The objective of the problem can thus be expressed as:

$$\mathbf{P0} : \max_{\{a(t)\}} (500 - T_{max}) \cdot I + \sum_{t=0}^{T_{max}} R_{i^*(t)}(t) \Delta t / C_T \quad (5)$$

$$\text{s.t. } \|r(t+1) - r(t)\| \leq V_{max}, \quad (6)$$

$$r(t) \in \mathcal{G}, \quad \forall t. \quad (7)$$

$$r(T_{max}) = r(0) \quad (8)$$

$$0 \leq T_{max} \leq 500 \quad (9)$$

$$\sum_t (R_{i^*(t)}(t) \Delta t) < C_i \quad \forall i \quad (10)$$

$$(11)$$

Where T_{max} is the time when all data have been downloaded or the UAV is landed. The total score is the objective function as defined in the AADM challenge [1], and consists of two terms: $F_a(\cdot) = S_1 + S_2$ where $S_1 = (500 - T_{max})$ is the score awarded if all data from every BS is successfully downloaded, which is specified using the indicator function I

$$I = \begin{cases} 1, & \text{if } \sum_t (R_{i^*(t)}(t) \Delta t) = C_T, \\ 0, & \text{O.W.} \end{cases}$$

S_2 is the total amount of data downloaded from all BSs, normalized to the sum of data available in BSs.

Constraint 10 specifies that the UAV cannot download more than the demand of each BS from any BS.

Notice that this formulation assumes the UAV will successfully get back to its initial position and land; For completion, a different version of the problem considering the 20% penalty of not returning should be solved, and the solution with maximum score out of the two versions will be returned, due to mutual exclusivity, this will be optimal.

III. BASELINE METHOD

The baseline development focused on comparing the effect of BS selection strategy on the achieved score. We tested Greedy vs Optimal BS selection, creating a practical reference strategies for trajectory control and BS selection.

A. Fixed Trajectory

In this approach, the UAV follows a predefined path composed of deterministic waypoints, $\mathcal{W}_{fx}$, that ensure geofence compliance, i.e. $\forall w \in \mathcal{W}_{fx}, w \in \mathcal{G}$. During flight, the UAV continuously monitors the instantaneous SNR values $\gamma_i(t)$ reported by each BS.

B. Base Station Selection

1) *Greedy*: In this BS selection strategy, the UAV always connects to the BS with the highest SNR at that time:

$$i^*(t) = \arg \max_i \gamma_i(t). \quad (12)$$

This method provides a simple yet effective benchmark by always maintaining the best instantaneous link quality without dynamically altering the trajectory. Formally, the UAV chooses its location from $\mathcal{W}_{fx}$.

$$p_{t+1} = w_{t+1}, \quad (13)$$

where w_{t+1} is the $t+1$ 'th element in $\mathcal{W}_{fx}$.

2) *Optimal BS selection*: In this BS selection strategy, assuming a fixed trajectory, we formulate the BS selection problem as an MILP, and find the optimal selection. From the fixed trajectory, we can calculate the SNR that the UAV will experience from each BS, hence, the time-series of potential data rates $R_i(t)$ is known for all BS_i . The goal is to find the optimal BS selection $i^*(t)$ at each time step t to maximize the total score as given in 5.

Notice that the objective is non-linear due to the indicator function I but for i) the fixed trajectory setting and ii) predefined data demand, this term does not influence the algorithm; By simply maximizing for the total downloaded data rate, the optimal solution will be obtained.

We introduce a binary decision variable $x_i(t) \forall i, \forall t$ which is 1 if BS i is selected at time t , 0 otherwise. We have:

$$\sum_i x_{i,t} = 1, \quad \forall t \in \mathcal{T}. \quad (14)$$

to ensure exactly one BS is selected at each time. Assuming the known waypoints and the binary variables $x_i(t)$, problem $\mathbf{P0}$ can be solved as an ILP [2].

IV. PROPOSED METHODOLOGY

In this section, we present our adaptive trajectory planning and base-station (BS) selection framework for time-limited data collection in the AERPAW digital twin environment. Building on the system model described earlier, the key idea is to jointly optimize (i) the visiting order of BS locations and (ii) the BS association along the resulting trajectory. Additionally, we perform a binary search on parameter T_{Max} , total time of flight for the UAV, such that minimum time which results in the maximum score for the experiment is calculated.

The UAV begins its mission near LW1 and must collect data from four BSs located at $\{LW1, LW2, LW3, LW4\}$. Our method first assumes $T_{Max} = 500s$, then enumerates all possible permutations of visiting the remaining three BS locations, constructs a feasible waypoint sequence for each candidate order, and then solves a Mixed Integer Linear Program (MILP) to compute the optimal BS-selection schedule for that trajectory. The final trajectory is selected as the one achieving the maximum AADM score. If the UAV was able to download all data from all BSs with the current value of T_{max} , then we will repeat the process in a binary search matter for

another $0 < T_{Max} < 500s$ value until the smallest such T_{Max} is found.

A. Adaptive Trajectory Construction

Let $\mathcal{L} = \{LW1, LW2, LW3, LW4\}$ denote the BS locations. Since the UAV starts near LW1 and also visits it on its way back for landing, we consider all permutations of the other three locations:

$$\Pi = \{\pi = (\pi_1, \pi_2, \pi_3) \mid \{\pi_1, \pi_2, \pi_3\} = \{LW2, LW3, LW4\}\}.$$

For each $\pi \in \Pi$, we generate a deterministic waypoint sequence

$$\mathcal{W}(\pi) = [w_0, w_1, \dots, w_{T(\pi)}],$$

where w_0 is the starting waypoint near LW1 and subsequent waypoints guide the UAV sequentially through $\pi_1 \rightarrow \pi_2 \rightarrow \pi_3$.

Each trajectory is formed by forming the straight-line path between consecutive LW locations while ensuring both the UAV's maximum-velocity constraint

$$\|r(t+1) - r(t)\| \leq V_{\max}, \quad (15)$$

and the geofence constraint

$$r(t) \in \mathcal{G}, \quad \forall t. \quad (16)$$

1) *Intermediate Waypoint for LW3*: Direct straight-line motion to LW3 may violate the geofence for certain departure points (e.g., from LW2 or LW4). To enforce geofence compliance, when a segment of the trajectory is intended to reach LW3, we introduce an intermediate "middle point" w_{mid} :

- w_{mid} is positioned safely inside $\mathcal{G}$ along a feasible corridor between the current waypoint and LW3.
- The trajectory segment is then constructed as $w_{\text{curr}} \rightarrow w_{\text{mid}} \rightarrow LW3$.

This two-hop waypoint guarantees that all intermediate waypoints satisfy geofence and velocity constraints.

2) *Adaptive Speed Adjustment Near Target BS*: To increase the time the UAV spends within high-SNR regions around each BS, we introduce an adaptive-speed mechanism:

- When the UAV is within a predefined radius of the next LW location, its speed is reduced to $V_{\max}/2$.
- Otherwise, the UAV travels at $V_{\max}$.

This slowdown near BSs increases achievable throughput by allowing the UAV to take advantage of stronger channel conditions for longer durations. The waypoint-generation procedure integrates this by adjusting the time-step spacing while preserving constraints on inter-waypoint distances.

B. MILP-Based BS Selection Along a Given Trajectory

For each trajectory candidate $\mathcal{W}(\pi)$, the UAV's position $r(t)$ at all time steps is predetermined. Therefore, the instantaneous SNR from BS i ,

$$\gamma_i(t) = \frac{P_i G_i}{N_0 L_i(r(t))},$$

and resulting downlink rate $R_i(t) = f(\gamma_i(t))$, can be computed directly using the channel model which is free space path loss.

Let $x_{i,t} \in \{0, 1\}$ be a binary decision variable indicating the selected BS at time step t . The UAV must be connected to exactly one BS:

$$\sum_i x_{i,t} = 1, \quad \forall t. \quad (17)$$

The total amount of data downloaded from BS i is:

$$D_i = \sum_t x_{i,t} R_i(t) \Delta t, \quad (18)$$

and must satisfy the demand constraint:

$$D_i \leq C_i, \quad \forall i. \quad (19)$$

Since, for fixed trajectories, maximizing the total downloaded data is equivalent to maximizing the AADM score, the MILP solves:

$$\max_{x_{i,t}} \sum_i \sum_t x_{i,t} R_i(t) \Delta t, \quad (20)$$

subject to the constraints above. Gurobi is used as the MILP solver to obtain optimal BS association variables $x_{i,t}^*$. The active BS at time t is then:

$$i^*(t) = \arg \max_i x_{i,t}^*. \quad (21)$$

C. Grid Search Over All Candidate Trajectories

Each visiting order π is evaluated by:

- 1) Constructing $\mathcal{W}(\pi)$ with geofence compliance, middle waypoint insertion (if needed), and adaptive-speed control.
- 2) Computing the per-BS rate profiles $\{R_i(t)\}$.
- 3) Solving the MILP to obtain the optimal BS-selection schedule $\{i^*(t)\}$.
- 4) Computing the resulting AADM score $S(\pi)$.

The optimal visiting order and trajectory are:

$$\pi^* = \arg \max_{\pi \in \Pi} S(\pi). \quad (22)$$

Algorithm 1 illustrates the operation of **Adaptive Trajectory and BS Selection**.

V. EXPERIMENTAL EVALUATION

In this section, we evaluate the effectiveness of the proposed adaptive trajectory planning and MILP-based BS-selection framework. We compare the performance of the optimized trajectory against the fixed-trajectory baseline developed in earlier stages of the project. Our evaluation consists of four components:

- 1) Trajectory visualization under different visiting orders.
- 2) Score comparison between fixed and optimized trajectories in different total data demand values.
- 3) Instantaneous SNR behavior along optimized trajectories.
- 4) Statistical analysis of permutation frequencies under repeated trials.

Comparison: Fixed vs. Optimized Trajectory (Variable Speed)

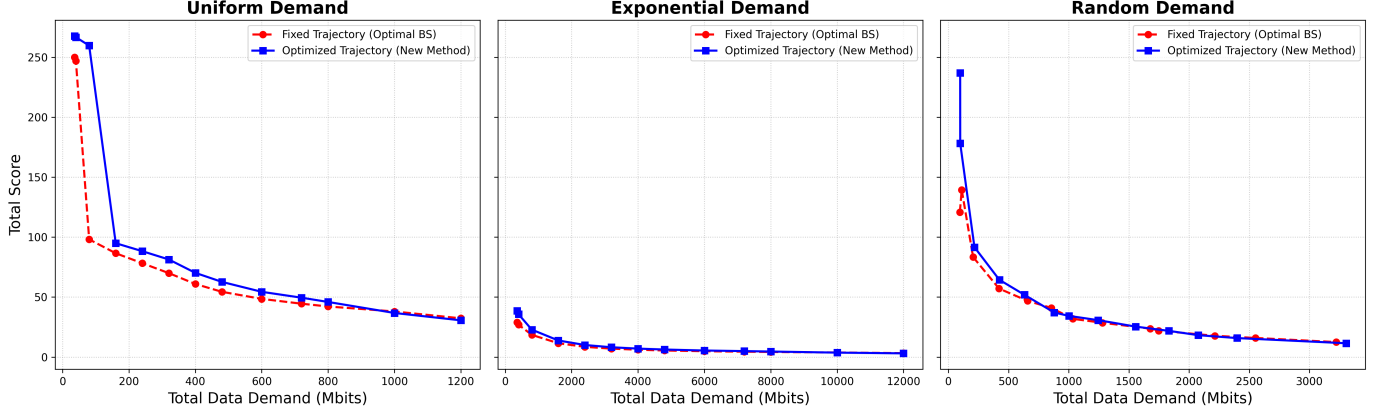


Fig. 1: Score comparison under Uniform, Exponential, and Random demand distributions for fixed vs. optimized trajectories.

Algorithm 1 Adaptive Trajectory and BS-Selection Optimization

Input: BS locations $\mathcal{L}$, geofence $\mathcal{G}$, $V_{\max}$, Δt , demands $\{C_i\}$.

Output: Optimal trajectory $\mathcal{W}(\pi^*)$ and BS schedule $\{i^*(t)\}$.

- 1: Initialize w_0 near LW1.
- 2: Generate all permutations Π of $\{\text{LW2, LW3, LW4}\}$.
- 3: **for** each $\pi \in \Pi$ **do**
- 4: Construct $\mathcal{W}(\pi)$:
 - Discretize the path between successive LW locations.
 - If traveling to LW3, insert w_{mid} and generate a two-segment path.
 - Ensure all waypoints satisfy geofence and velocity constraints.
 - Apply adaptive-speed control: use $V_{\max}/2$ when near the target LW, otherwise use $V_{\max}$.
- 5: For all t and i , compute $\gamma_i(t)$ and $R_i(t)$.
- 6: Formulate and solve the MILP:
 - Decision variables: $x_{i,t} \in \{0, 1\}$.
 - Constraints: $\sum_i x_{i,t} = 1$, $\sum_t x_{i,t} R_i(t) \Delta t \leq C_i$.
 - Objective: maximize $\sum_i \sum_t x_{i,t} R_i(t) \Delta t$.
- 7: Extract $i^*(t)$ and compute score $S(\pi)$.
- 8: **end for**
- 9: $\pi^* \leftarrow \arg \max_{\pi} S(\pi)$.
- 10: **return** $\mathcal{W}(\pi^*)$ and $\{i^*(t)\}$.

Unless otherwise stated, all experiments assume a total available mission time of 500 seconds, the AERPAW geofence boundaries, and the same UAV movement constraints used in prior stages.

A. Trajectory Visualization

1) *Fixed Trajectory Baseline:* Figure 2 shows the fixed trajectory used as the baseline in the earlier stages of the project. The UAV follows a predetermined closed-loop path inside the

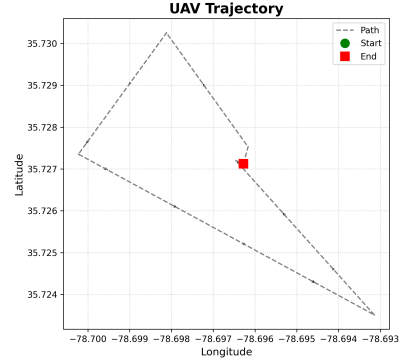


Fig. 2: Fixed UAV trajectory used as baseline.

geofence, starting and ending near LW1. BS selection was performed using MILP while the trajectory remained fixed.

2) *Optimized Trajectories Across Permutations:* Among the 6 possible permutation of visiting LW2, LW3, and LW4, only 4 of the permutations is optimal across different data demand scenarios. In this section, we present the 4 candidate UAV trajectory when operating based on **Adaptive Trajectory and BS Selection** when the data demand for each BS is 100. Figures 3a–3d illustrate the optimized trajectories obtained for each permutation.

B. Score Comparison

To demonstrate the performance gain achieved via trajectory optimization, we compare the AADM scores between:

- **Fixed Trajectory + Optimal BS Selection (Baseline)**
- **Optimized Trajectory + MILP BS Selection (Proposed)**

For a more generalized comparison and demonstrating the robustness of our methodology, we considered three different data demand scenarios, i.e.e, Uniform, Exponential, and Random. Figure 1 summarizes the results and shows consistent gain of our proposed methodology over the baseline. In our

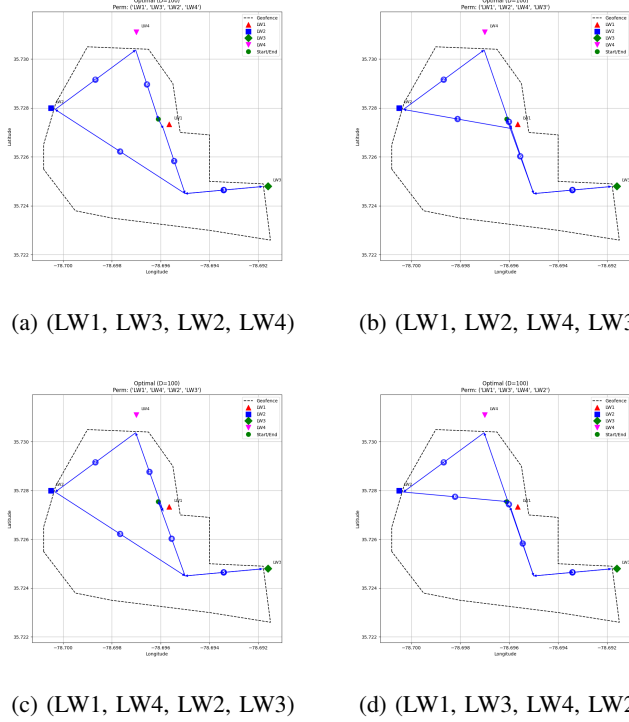


Fig. 3: Different optimized trajectories with different LW visiting orders.

previous report, we explicitly showed that performance of simply integrating fixed trajectory with greedy BS selection (BS with the highest SNR) is lower than employing our adaptive BS selection.

C. Instantaneous SNR Analysis

To illustrate how UAV adaptive trajectory and optimal assignment to different BSs affect channel quality, we examine the instantaneous SNR profiles along the optimized trajectories. These profiles determine the achievable download rate and strongly effect the MILP-based BS selection.

Figure 4 presents the SNR variations for all four representative trajectory permutations.

D. Permutation Frequency Analysis

To understand which visiting order tends to be optimal in a general perspective, we execute the full optimization pipeline across large number of randomized data-demand scenarios and record how often each permutation corresponds to the optimal trajectory. Table I illustrates the selection frequency of different permutations over large number of simulations.

TABLE I: Frequency of Optimal Visiting Order Across Randomized Trials

Permutation	Count	Frequency (%)
(LW1, LW2, LW4, LW3)	231	55.5%
(LW1, LW3, LW4, LW2)	93	22.4%
(LW1, LW3, LW2, LW4)	49	11.8%
(LW1, LW4, LW2, LW3)	43	10.3%

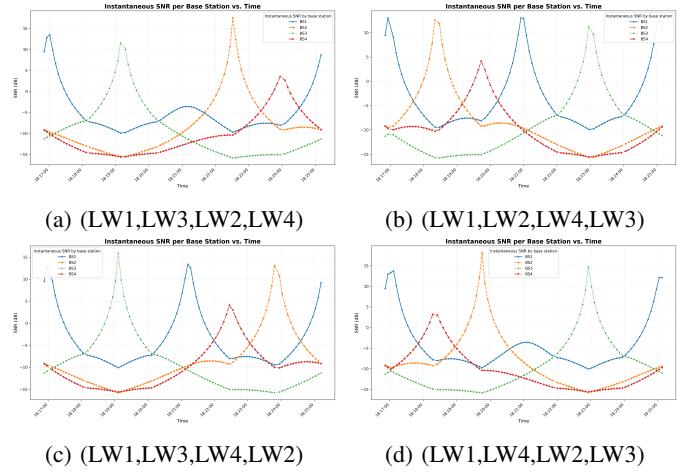


Fig. 4: Instantaneous SNR of each BS for all four trajectory permutations

VI. CONCLUSION AND FUTURE WORKS

In this project, we developed an adaptive trajectory optimization framework for UAV-based data collection in LTE/5G environments. Our methodology operates based on evaluating all feasible visiting orders of base stations, while respecting the geofence constraint with adaptive speed control, and applying MILP-based BS selection along each trajectory. The proposed method consistently outperformed the fixed-trajectory baseline across multiple data demand scenarios. The optimized paths generates higher AADM scores and more favorable SNR conditions, demonstrating the benefit of jointly considering trajectory and link quality.

Future works in this direction can involve deploying more UAVs for data downloading and jointly optimizing their trajectories via learning-based optimization methodologies and comparing with state-of-the-art methods.

VII. CONTRIBUTION

- 1) Reza has worked on the Fixed-Trajectory section.
- 2) Farzad has worked on the optimal BS selection section.
- 3) Both equally worked on developing the adaptive trajectory optimization framework.
- 4) Both equally contributed to the rest of the project, including report, setting up the experiments, and project planning.

REFERENCES

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- [2] Gurobi Optimization, LLC, "Gurobi Optimizer Reference Manual," 2024. [Online]. Available: <https://www.gurobi.com>